

Topic: CO1**Security Features in Real-World Applications**

1. Identify any five real-world applications or online services that involve communication, authentication, or transactions. For each application, investigate and identify:

Application	Security Feature	Security Mechanism	Protocol / Technology	How it provides security
Instagram	Authentication	Password and Multifactor authentication	OAuth 2.0, TLS	Verifies user's identity and prevents Unauthorized access
WhatsApp	Confidentiality	End to End Encryption	Signal Protocol	It converts messages into unreadable form so unauthorized people cannot understand it while it is being sent
Google Drive	Authorization	Access control or permissions	OAuth 2.0 , TLS	Allows the owner to decide who can view, edit and share a file. It prevents unauthorized access.
YouTube	Availability	Redundant servers and load balancing	cloud infrastructure	It uses multiple servers to handle requests from user and keep the service available even if server fails
Google pay	Integrity	Cryptographic Protection	TLS	Protects payment information and prevents unauthorized modification of transaction data

2. For any five real-world applications, investigate the security controls used to protect users and data. Classify the controls based on whether they provide Confidentiality, Integrity, or Availability, and determine how these controls help in preventing, detecting, and recovering from security attacks. Mention the mechanisms and protocols involved.

Application	Security Control	CIA Classification	Prevention	Detection	Recovery	Mechanism and Protocol
Instagram	Password and Multifactor Authentication	Confidentiality	MFA makes it difficult for someone to enter the account even if they know the password	Login alerts can notify the user about an unknown or suspicious login	The user can change the password and use account recovery if the account is hacked	OAuth 2.0, TLS, MFA
WhatsApp	End-to-End Encryption	Confidentiality	Messages are encrypted so that other people cannot read them while they are being sent	Security notifications can show when the security code of a chat changes	The account can be verified again and backups can help restore chats when supported	Signal Protocol, TLS

Google Drive	File Permissions and Access Control	Confidentiality and Integrity	The owner can decide who can view or edit a file, which reduces unauthorized access or changes	File activity and notifications can help the user notice unwanted changes or access	Version history allows an older version of a file to be restored if something goes wrong	OAuth 2.0, TLS, Access Control
YouTube	Multiple Servers and Load Balancing	Availability	Requests are shared between different servers so one server failure does not easily stop the service	The system can monitor server problems and unusual traffic	Backup and alternate servers can keep the service running when a server fails	HTTPS/TLS, Cloud Infrastructure and Load Balancing
Google Pay	Encryption and Transaction Authentication	Integrity and Confidentiality	Encryption protects payment information and authentication helps prevent unauthorized transactions.	The system monitors transactions and alerts the user when it notices suspicious or unusual payments	The user can report unauthorized transactions and take steps to secure the account and resolve the issue	TLS, Encryption, Tokenization, Authentication